

Real-Time Vibration and Temperature Monitoring System Using STM32 and MPU6050

Gundu Naga Sowmya¹, Payakattu Lahari², Surekha Paneerselvam³, V. S. Kirthika Devi⁴

Department of Electrical and Electronics Engineering

Amrita School of Engineering, Bengaluru, Amrita Vishwa Vidyapeetham, India

Email: bl.en.p2ebs25007@bl.students.amrita.edu¹, bl.en.p2ebs25013@bl.students.amrita.edu²,
p_surekha@blr.amrita.edu³, s_kirthika@blr.amrita.edu⁴

Abstract—This study describes a real-time vibration and temperature monitoring system that uses an analog temperature sensor, an MPU6050 MEMS accelerometer, and an STM32 microprocessor. To track machine operating conditions, the suggested system continuously collects heat readings and tri-axial vibration data. The sensor data is processed to obtain time-domain statistical features for condition assessment, including mean, root mean square (RMS), standard deviation, and peak-to-peak values. To differentiate between problematic and normal operational states, a machine-learning-based categorization approach is utilized. TinyML is used to install the learned model on the STM32 microcontroller, allowing for entirely edge-based real-time inference independent of the cloud. Experimental results reveal that the system reliably detects vibration disturbances and anomalous temperature rise, enabling prompt problem indication through LED alerts and serial monitoring. The suggested technology offers a low-cost, low-latency, and efficient approach for embedded condition monitoring and predictive maintenance applications.

Index Terms—STM32 microcontroller, MPU6050, MEMS accelerometer, temperature monitoring, vibration analysis, feature extraction, machine learning, predictive maintenance, embedded system, real-time monitoring.

I. INTRODUCTION

The increasing demand for reliable machine-health monitoring has accelerated the development of low-cost, embedded predictive-maintenance systems. Mechanical failures in small- and medium-scale machines often originate from excessive vibration, bearing wear, imbalance, or temperature increase, which are difficult to detect through periodic manual inspection. Recent studies highlight that vibration signals measured through MEMS-based accelerometers provide highly sensitive indicators of early-stage mechanical degradation, enabling accurate fault detection before catastrophic failure occurs [1]–[4].

Changes in temperature add to changes in vibration, giving more information about friction, overload, and thermal stress. This makes condition-monitoring systems more reliable. Real-time temperature monitoring is crucial for maintaining safety in challenging industrial settings, according to recent studies. Krishna et al. demonstrated how continuous sensing and real-time data processing may greatly improve early hazard detection and operational reliability by proposing an integrated airflow and temperature monitoring system for underground mine safety. This demonstrates the increasing demand for

embedded, sensor-based monitoring systems that can function reliably and with minimal delay in real-world circumstances [5]–[6].

On-device sensing, preprocessing, and lightweight inference for real-time diagnostics have been made possible by developments in embedded microcontrollers, such as the ARM Cortex-M and STM32 platforms. Numerous studies claim that contemporary MCUs effectively manage statistical feature extraction, signal transformation, and multi-axis vibration data collection while preserving low latency and low power consumption [7]–[8].

MEMS sensors, like the MPU6050, are widely used for monitoring rotating equipment, gearboxes, motors, and structural components because of their small size, low cost, and capacity to record tri-axial vibration signatures with high resolution [9]–[10].

Additionally, research shows that combining temperature and vibration monitoring improves decision-making accuracy, especially when identifying problems linked to anomalous thermal and dynamic behavior [11]–[12].

In embedded systems, machine-learning-based fault classification has gained significant attention in recent years. Several studies in the literature have demonstrated that statistical analysis of vibration signals combined with machine learning classifiers such as decision trees, support vector machines, and shallow neural networks can effectively identify abnormal mechanical behavior in rotating and stationary systems [13], [14]. Prior research has also explored advanced signal analysis techniques to improve fault detection accuracy under complex operating conditions [15]. These developments collectively reflect the growing trend toward deploying lightweight, TinyML-based models for reliable edge-level diagnostics on resource-constrained microcontrollers.

In settings where direct human involvement is challenging, real-time temperature monitoring and control technologies are crucial. Closed-loop control and continuous sensing enhance system stability and operational safety [16].

Real-time data collection and processing from several sensors is made possible by embedded multi-sensor monitoring systems, which enhance system dependability and early defect identification. These systems are appropriate for smart, industrial, and real-time monitoring applications because they provide continuous monitoring of temperature and ambient

data when paired with IoT-based platforms and anomaly detection algorithms [17]-[18].

Motivated by these results, the suggested system uses an analog temperature sensor, an MPU6050 MEMS accelerometer, and an STM32 microcontroller to provide a real-time vibration and temperature monitoring solution. On-device statistical feature extraction and threshold-based decision logic and embedded machine-learning inference are used to detect anomalous machine behavior. This design is in line with current research trends that highlight affordable, scalable, and on-device predictive-maintenance frameworks appropriate for industrial settings and small-scale machines.

II. PROPOSED SYSTEM

The proposed system integrates vibration and temperature sensing on an STM32 microcontroller architecture to offer a low-cost, real-time condition-monitoring solution. The MPU6050 MEMS accelerometer is used to continually measure a machine's mechanical behavior, while a temperature sensor is used to simultaneously record thermal fluctuations. The solution avoids the need for expensive analyzers or cloud-based computing frameworks by carrying out feature extraction and decision-making locally on the microcontroller. The architecture of the proposed system is shown in Fig. 1.

The I2C protocol is used to interface the MPU6050 to the STM32 in order to gather tri-axial acceleration data at a predetermined sampling rate. After preprocessing, these unprocessed signals are transformed into useful vibration indicators. To depict the intensity of vibrations, important time-domain characteristics like mean, RMS, standard deviation, and peak-to-peak value are calculated. To characterize vibration behavior linked to imbalance or mechanical wear, time-domain statistical features are computed from the sensor data. In order to monitor thermal behavior and ensure that overheated failures are also recorded, an analog temperature sensor is interfaced to the STM32 ADC in parallel.

To categorize the machine state as "Normal" or "Faulty," a simple decision logic is constructed on the STM32. A TinyML-based model trained on extracted features or a threshold-based detection method based on empirically recorded baseline values may be used for this categorization. When an aberrant state is detected, the STM32 sends processed data to a serial terminal (PUTTY) for external visualization and logging, as well as visual alarms via red and green LEDs. This guarantees real-time feedback, allowing operators to react promptly to new errors.

Because of its autonomous operation and low power consumption, the system can be used for instructional displays, laboratory setups, and small-scale industrial machinery. The suggested solution delivers dependable, quick, and economical predictive maintenance without requiring complicated hardware or cloud connectivity by combining multi-sensor fusion, embedded feature extraction, and on-device analytics.

III. METHODOLOGY

Sensor acquisition, signal preprocessing, feature extraction, classification, and warning creation comprise the or-

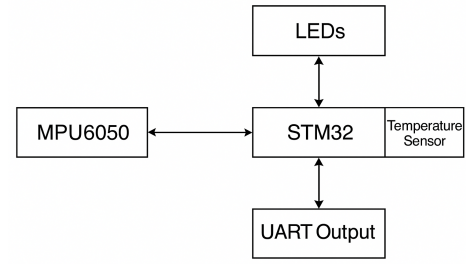


Fig. 1. Hardware architecture of the proposed STM32-based vibration and temperature monitoring system

ganized sequence of the approach used in this work. The hardware–software integration of the MPU6050 accelerometer, temperature sensor, serial monitoring environment, and STM32 microcontroller is illustrated in Fig. 1.

1) *Vibration Sensor, MPU6050*: The MPU6050 is a 6-axis MEMS accelerometer and gyroscope that measures the machine's vibration levels in real time. It offers X, Y, and Z-axis acceleration data that aids in the identification of mechanical flaws, imbalance, and misalignment. To ensure quick and precise measurements, the sensor uses the I2C protocol to transmit digital vibration data to the STM32.

2) *Temperature Sensor*: The machine's thermal state is tracked by the temperature sensor. It generates a temperature-proportional analog output that is sent to the ADC pin of the STM32. When vibration is insufficient to detect overheating, friction-based heating, or thermal defects, this block can aid.

3) *STM32 Microcontroller*: The STM32 is the core processing unit of the system. It performs the following operations:

- Reads vibration data from MPU6050 via I2C
- Reads temperature data via ADC
- Filters and preprocesses the raw signals
- Extracts important features (RMS, Mean, Std, P2P)
- Compares features with threshold values
- Classifies machine condition as Normal or Faulty
- Controls LED indicators
- Sends data to the PC via UART

The STM32 ensures real-time, low-latency processing without requiring a PC or cloud.

4) *LED Indicators*: The STM32 GPIO pins are linked to two LEDs:

- Green LED → Typical Situation
- Red LED → Error Situation

These LEDs are controlled by the STM32 using temperature and vibration decisions. This eliminates the requirement for a display and provides the operator with instant visual feedback.

5) *UART Output (Serial Monitor)*: A PC terminal (PUTTY/Serial Monitor) receives processed vibration values, temperature data, and fault states from the STM32. For real-time data streaming, UART is utilized because it is dependable and easy to use

6) *Getting data and setting up a system:* The suggested system uses an STM32 microcontroller with a motion sensor and a temperature sensing unit to monitor vibration and temperature in real time. Sensor data is continuously captured and transferred via UART to a host system throughout the training phase. The obtained data represents both normal operating conditions and fault circumstances, including vibration, tilt disturbances, and anomalous temperature rise. During deployment, all sensing, processing, and decision-making are executed locally on the STM32 microcontroller, providing a fully edge-based predictive maintenance system.

7) *Dataset Generation:* A Python-based serial interface is used to log real-time sensor values into a structured CSV dataset. Each data record consists of the following parameters:

- ADC value in its raw form
- Voltage conversion
- Degrees Celsius of temperature
- condition of vibration detection
- Time stamp

8) *Data Preprocessing:* Before the model is trained, the gathered dataset is preprocessed to enhance data quality and model resilience. This includes:

- Removal of insufficient and noisy samples
- Normalization for temperatures and voltage measurements
- Segmenting sensor data into time periods with predetermined durations

9) *Feature Extraction:* The underlying system behavior is captured by extracting a set of time-domain and frequency-domain properties from each time window. Among the traits that were extracted are:

- Mean
- Standard deviation
- RMS, or root mean square
- Value from peak to peak
- Deviation in temperature
- Temperature change rate

A. Mathematical Formulation

Let x_i represent the sampled vibration signal, where $i = 1, 2, \dots, N$ and N is the total number of samples.

Mean:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

Root Mean Square (RMS):

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (2)$$

Standard Deviation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (3)$$

Peak-to-Peak Value:

$$P2P = x_{\max} - x_{\min} \quad (4)$$

Temperature Conversion (ADC to °C):

$$T(^{\circ}C) = \frac{V_{\text{out}} - V_{\text{offset}}}{\text{Sensitivity}} \quad (5)$$

These characteristics offer a condensed and clear depiction of the vibration and thermal properties needed for accurate fault classification.

1) *Machine Learning Model Training:* The extracted feature set is used to assess a number of supervised machine learning techniques, such as Random Forest Classifier, Support Vector Machine (SVM), Logistic Regression, and a lightweight Neural Network. Classification accuracy, computational complexity, and embedded deployment appropriateness are used to assess the model's performance. The Random Forest classifier outperformed the other models under evaluation in terms of classification performance and computational cost. It was therefore chosen to be implemented in the suggested system.

2) *TinyML Deployment and Real-Time Inference:* The STM32 firmware incorporates the taught machine learning model as a C-array after converting it into TensorFlow Lite Micro format. The microcontroller carries out the following actions when operating in real time:

- Constant collection of temperature and vibration data
- Feature extraction using windows
- On-device machine learning inference
- System state classification as either normal or faulty

LED indicators are used to provide visual feedback based on the inference outcome. Without requiring cloud connectivity, this TinyML-based method allows for effective and dependable defect detection directly on an ARM Cortex-M microcontroller with limited resources.

B. SOFTWARE IMPLEMENTATION

STM32CubeIDE is used in the development of the firmware. The implementation of the proposed approach is illustrated in Fig. 2. STM32CubeMX is used to configure peripheral modules such as I2C, ADC, UART, GPIO, and timer interrupts. Vibration and temperature signals are periodically sampled by the microcontroller, which then preprocesses the data to eliminate noise and normalize it. Each acquisition window is used to extract time-domain features such as mean, RMS, standard deviation, and peak-to-peak value. Statistical feature analysis is used for vibration condition assessment, to find anomalous spectral patterns. Machine conditions are categorized as either Normal or Faulty using a threshold-based decision method. LED indicators and serial messages are used to convey corresponding notifications. interface to a computer for real-time visualization.

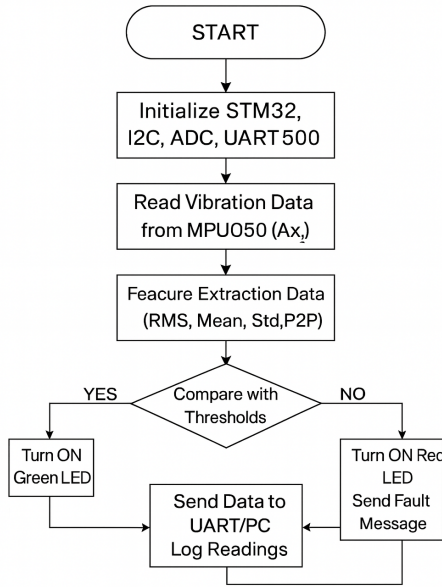


Fig. 2. Flowchart of the proposed system

C. HARDWARE IMPLEMENTATION

The MPU6050 MEMS accelerometer for tri-axial vibration measurement, an analog temperature sensor for thermal monitoring, and the STM32 microcontroller as the central processing unit make up the hardware architecture. Continuous acceleration data acquisition is made possible by the MPU6050's link with the STM32 via the I2C communication channel. To measure temperature changes, the ADC module is attached to the temperature sensor. The system status is shown by two LEDs: a red LED for fault circumstances and a green LED for normal operation. Processed data is sent via a serial terminal via a UART interface to a computer for real-time visualization.

IV. EXPERIMENTAL ANALYSIS

The entire hardware configuration for the suggested system is depicted in this diagram. The MPU6050 vibration sensor, temperature sensor, indicator LEDs, and UART module are all interfaced with the STM32 microcontroller. To make testing, signal monitoring, and real-time data collection easier, every component is installed on a breadboard.

The green LED turns ON when both vibration and temperature values remain within the predefined normal range. This indicates that the machine is operating under stable conditions without abnormal disturbances. The UART output during this state shows low RMS and stable temperature readings.

During anomalous vibration or when the temperature values exceed threshold limits, the red LED is instantly turned on by the STM32 upon detecting the deviation. In this scenario, the UART output confirms the existence of a defect by displaying high RMS, peak-to-peak vibration values, or elevated temperature. Sample sensor values along with ML labels are presented in Table I.

```

... SVM
Accuracy : 83.3
Precision: 66.7
Recall   : 66.7
F1 Score : 66.7

... LOGISTIC REGRESSION
Accuracy : 72.2
Precision: 50.0
Recall   : 66.7
F1 Score : 55.6

... Accuracy : 83.3
Precision: 83.3
Recall   : 100.0
F1 Score : 88.9

... MODEL PERFORMANCE COMPARISON (Stratified K-Fold)
Random Forest
Accuracy : 83.3
Precision: 83.3
Recall   : 100.0
F1-Score : 88.9
-----
Logistic Regression
Accuracy : 72.2
Precision: 50.0
Recall   : 66.7
F1-Score : 55.6
-----
SVM
Accuracy : 83.3
Precision: 66.7
Recall   : 66.7
F1-Score : 66.7
-----
  
```

Fig. 3. Console output showing comparative performance evaluation of Random Forest, SVM, and Logistic Regression classifiers

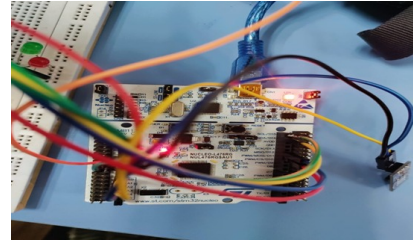


Fig. 4. Hardware setup of the STM32-based vibration and temperature monitoring system

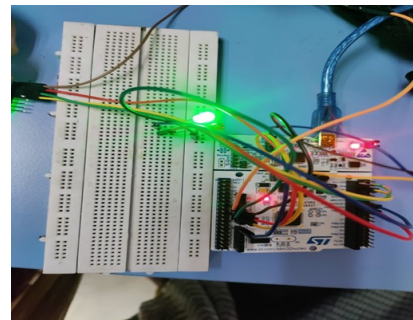


Fig. 5. Normal operating condition indicated by the green LED on the STM32 board

TABLE I
SAMPLE SENSOR DATA AND ML LABELS

Sample	Temperature	Vibration	RMS value	Class Label
1	24.1	No	0.12	Normal
2	13.3	Yes	0.46	Fault
3	47.8	No	0.19	Fault
4	12.0	Yes	0.51	Fault

TABLE II
MACHINE LEARNING MODEL PERFORMANCE COMPARISON

Model	Acc.	Prec.	Rec.	F1
Random Forest	83.3	83.3	100.0	88.9
SVM	83.3	66.7	66.7	66.7
Logistic Reg.	72.2	50.0	66.7	55.6

TABLE III
REAL-TIME INFERENCE RESULTS ON STM32

Operating Condition	ML Output	LED Indication
Normal Temperature	Normal	Green
Vibration Detected	Fault	Red
High Temperature	Fault	Red
Stable Operation	Normal	Green

V. RESULTS AND DISCUSSION

An analog temperature sensor, an MPU6050 accelerometer, and an STM32 microcontroller were used in an experimental evaluation of the suggested real-time vibration and temperature monitoring system. To ensure accuracy, responsiveness, and dependability, the system was tested under various operating situations. The hardware configuration and system behavior in both normal and malfunctioning machine states are shown in Fig. 4 and Fig. 5 respectively.

The MPU6050 generated low and constant acceleration values in all three dimensions when operating steadily and without outside interference. RMS, standard deviation, and peak-to-peak amplitude were among the recovered time-domain parameters that stayed within the predetermined threshold ranges. Additionally, consistent readings around ambient levels were observed by the temperature sensor. As seen in Fig. 4, the STM32 properly identified the machine state as Normal in this scenario and turned on the green LED. Stable readings with little variation were confirmed by the UART output.

From Table II, it is observed that the Random Forest classifier achieves higher recall and F1-score compared to Logistic Regression and SVM, indicating superior fault-detection reliability for the proposed application.

The system was subjected to controlled disturbances like tapping, shaking, and minor tilting in order to validate vibration-based problem detection. Significant increases in vibration amplitude and changes in feature values were brought on by these disruptions. The fault threshold was triggered by a rapid increase in the RMS and peak-to-peak values. Consequently, as shown in Fig. 5, the STM32 changed the system state to Faulty, turning on the red LED. The applied disturbances were clearly represented by abrupt spikes in the vibration data on the UART output.

The temperature sensor was gradually heated in order to simulate a thermal malfunction. The ADC output grew proportionately when the temperature rose above the designated limit. Even if vibration levels stayed reasonable, the controller reported a failure when the threshold was exceeded. This shows that the temperature monitoring subsystem operates on its own and, when paired with vibration data, strengthens fault identification.

The technology was tested more thoroughly in scenarios where both temperature changes and vibration were present. The STM32 kept the system in the broken state by finding both deviations at the same time in these cases. This indicates that by minimizing false positives and delivering consistent classification under varied mechanical and environmental circumstances, multi-sensor fusion considerably boosts diagnostic accuracy.

When compared to single-sensor monitoring, the results reveal that integrating temperature and vibration features considerably boosts the accuracy of defect categorization. The Random Forest-based TinyML model performed within the STM32 microcontroller's memory and CPU limits while maintaining high accuracy.

During experimental testing, the system demonstrated a quick response time, reliable categorization, and no false positives. The results are presented in Tables II and III. These findings verify that the suggested method is appropriate for embedded predictive maintenance applications in real-time.

VI. CONCLUSION

The STM32 microcontroller, MPU6050 vibration sensor, and analog temperature sensor have all been successfully used in the design and implementation of the suggested real-time condition monitoring system. Accurate detection of heat and vibration-related anomalies was achieved using feature extraction, multi-sensor data fusion, and machine-learning-based classification. The MPU6050 successfully recognized variations in vibration strength, and the temperature sensor consistently recorded overheating situations, according to experimental data. Instantaneous and transparent system feedback was made possible by UART-based data logging and LED indicators.

The overall findings show that the system provides a dependable, economical, and efficient method for predictive maintenance in small-scale industrial and laboratory settings. The design does not require external hardware or rely on the cloud because all processing is done on the STM32 platform. To further increase classification accuracy and expand system capabilities, future improvements might use TinyML models and sophisticated signal processing.

REFERENCES

- [1] O. Oral, S. Bilgin, and M. U. Ak, "Evaluation of vibration signals measured by 3-axis MEMS accelerometer on human face using wavelet transform and classifications," *Tehnički Vjesnik – Technical Gazette*, vol. 29, no. 2, pp. 355–362, 2022.
- [2] I. A. Jamil, M. I. Abedin, D. K. Sarker, and J. Islam, "Vibration data acquisition and visualization system using MEMS accelerometer," in *Proc. Int. Conf. Electrical Engineering and Information and Communication Technology (ICEEICT)*, pp. 1–6, 2014.
- [3] M. K. Adeyeri, B. Mpofu, and B. Kareem, "Development of hardware system using temperature and vibration models integration concepts for machine monitoring: A case study," *Journal of Industrial Engineering International*, vol. 12, pp. 93–109, 2016.
- [4] J. Luo, Z. Wu, C. Shen, and F. Shen, "Development of real-time system for vibration detecting system," in *Proc. Int. Conf. Computer Application and System Modeling (ICCSM)*, 2010.
- [5] G. Merendino, A. Pieracci, M. Lanzoni, and B. Riccò, "An embedded system for real-time vibration analysis," in *Proc. IEEE Int. Workshop on Advances in Sensors and Interfaces (IWASI)*, pp. 6–11, 2011.

- [6] Krishna, JVN Sai, R. Anand, J. Ramprabhakar, V. P. Meena, and Francesco Benedetto. "Integrated airflow and temperature monitoring system for enhanced underground mine safety." In 2024 IEEE 3rd World Conference on Applied Intelligence and Computing (AIC), pp. 1235-1241. IEEE, 2024.
- [7] H. Zhang and W. Kang, "Design of the data acquisition system based on STM32," *Procedia Computer Science*, vol. 17, pp. 222–228, 2013.
- [8] Y. Lei, Z. He, and Y. Zi, "Condition monitoring and fault diagnosis of motor bearings using undersampled vibration signals from a wireless sensor network," *Journal of Sound and Vibration*, vol. 414, pp. 81–96, 2018.
- [9] A. Hodžić, "Advancing bevel gear assessment with a dedicated vibration sensor device," Master's thesis, University of Sarajevo, 2020.
- [10] R. B. Randall and J. Antoni, "Rolling element bearing diagnostics—A tutorial," *IEEE Signal Processing Magazine*, vol. 27, no. 6, pp. 18–30, Nov. 2010.
- [11] Y. Lei, Z. He, and Y. Zi, "A new approach to intelligent fault diagnosis of rotating machinery," *IEEE Transactions on Industrial Electronics*, vol. 55, no. 4, pp. 1593–1600, Apr. 2008.
- [12] J. P. Lynch and K. J. Loh, "A cost-effective and flexible vibration DAQ system for long-term continuous structural health monitoring," *Mechanical Systems and Signal Processing*, vol. 64–65, pp. 313–324, 2015.
- [13] S. Maldonado, A. Bravo, and R. Weber, "Vibration-based condition monitoring for rotating machinery using machine learning," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3452–3461, Jun. 2019.
- [14] M. Catelani, L. Ciani, and G. Patrizi, "Raspberry Pi as a low-cost data acquisition system for human-powered vehicles," *Measurement*, vol. 100, pp. 7–18, 2017.
- [15] J. Lee, H. Davari, J. Singh, and V. Pandhare, "Industrial artificial intelligence for smart manufacturing," *IEEE Industrial Electronics Magazine*, vol. 12, no. 2, pp. 18–28, Jun. 2018.
- [16] Paul, Rinika, and K. Deepa. "Closed-Loop AUV Temperature Control Using Internet of Underwater Things-Based Multi-Node Underwater Communication." In 2023 International Conference on Recent Advances in Information Technology for Sustainable Development (ICRAIS), pp. 49-53. IEEE, 2023.
- [17] Sahiram, Aswathi Hima, and Rajesh Kannan Megalingam. "Enhanced FPGA Based Multi-Sensor System for Health Monitoring with I2C Protocol." In 2025 8th International Conference on Circuit, Power Computing Technologies (ICCPCT), pp. 387-393. IEEE, 2025.
- [18] Abhiram, D., K. Dhanush Reddy, P. Koushik Reddy, and Shinu Rajagopal. "IoT Environmental Monitoring with Anomaly Detection." In 2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 699-704. IEEE, 2025.